

Soroush H. Zargarbashi

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About Me

PhD researcher in Trustworthy AI working on uncertainty quantification, robust/conformal prediction, with several top-tier ML papers (ICML, ICLR, NeurIPS) and industrial experiences on various topics from LLM fine-tuning, machine learning, and time-series forecast. Furthermore I have hands-on experience with software backend development.

Experiences

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| Microsoft , Applied Scientist Intern | Expected: Apr - Jul 2026 |
| <ul style="list-style-type: none"> Expected to begin in April 2026 | |
| Apple , Research Intern - Speech Translation Team | May - Oct 2025 |
| <ul style="list-style-type: none"> Finetuning LLMs for speech translation Experience with distributed multi-GPU training setups, data sharding techniques. Experience with speech embeddings, LLM finetuning, and data generation. | |
| CISPA Helmholtz Center for Information Security , Doctoral Researcher | Sep 2022 – Present |
| <ul style="list-style-type: none"> Conduct research on Trustworthy AI, focusing on uncertainty quantification, conformal prediction, and robustness. Developed efficient algorithms with provable coverage guarantees under adversarial perturbations. Published in top-tier venues including ICML 2024, ICLR 2025, and NeurIPS 2025. | |
| Dorfak Intelligent Trading - Quantitative Research , Senior Backend Developer | 2021 – 2022 |
| <ul style="list-style-type: none"> Designed and implemented microservice applications for scalable, intelligent trading systems, from data curation, strategy management, etc. Experienced in AI-powered and algorithmic trading for financial markets. Experienced in Django Rest Framework, Docker, etc. | |
| Aarman Pajouhan Sooba - Electricity market forecast , Senior Data and ML Scientist | 2020 – 2021 |
| <ul style="list-style-type: none"> Developed and maintained AI-based electricity market forecasting systems. Designed model fine-tuning and time series prediction modules; led research collaboration with international teams. | |

Publications

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| One Sample is Enough to Make Conformal Prediction Robust [Link to paper] 🔗
Soroush H. Zargarbashi , Mohammad Sadegh Akhondzadeh, Aleksandar Bojchevski | NeurIPS 2025 |
| Robust Conformal Prediction with a Single Binary Certificate [Link to paper] 🔗
Soroush H. Zargarbashi , Aleksandar Bojchevski | ICLR 2025 |
| Robust Yet Efficient Conformal Prediction Sets [Link to paper] 🔗
Soroush H. Zargarbashi , Mohammad Sadegh Akhondzadeh, Aleksandar Bojchevski | ICML 2024 |
| Conformal Inductive Graph Neural Networks [Link to paper] 🔗
Soroush H. Zargarbashi , Aleksandar Bojchevski | ICLR 2024 |

Conformal Prediction Sets for Graph Neural Networks [Link to paper] 	ICML 2023
<i>Soroush H. Zargarbashi, Simone Antonelli, Aleksandar Bojchevski</i>	
EvA: Evolutionary Attack on Graphs [Link to paper] 	ICLR 2026
<i>Sadegh Akhondzadeh⁺, Soroush H. Zargarbashi⁺, Jimin Cao, Aleksandar Bojchevski</i>	
Conformal Adaptive Test-time Scaling <i>Under review - ICML 2026 / Accepted ICLR 2026 workshop</i>	2026
<i>Mohammad Sadegh Akhondzadeh, Soroush H. Zargarbashi, Simone Antonelli, Aleksandar Bojchevski,</i>	
Front-Loaded Robust Conformal Prediction: Heavy Calibration and Light Weight Inference <i>Under review - ICML 2026</i>	2026
<i>Soroush H. Zargarbashi, Mohammad Sadegh Akhondzadeh, Aleksandar Bojchevski,</i>	
Quantifying Epistemic Predictive Uncertainty in Conformal Prediction <i>Under review - ICML 2026</i>	2026
<i>Siu Lun Chau, Soroush H. Zargarbashi, Yusuf Sale, Michele Caprio,</i>	
Efficient (Proxy) Robustness Certificates with Marginal Guarantee <i>Under preparation</i>	2026
<i>Soroush H. Zargarbashi⁺, Mohammad Sadegh Akhondzadeh⁺, Aleksandar Bojchevski,</i>	
Optimal Conformal Prediction under Epistemic Uncertainty [Link to paper] 	2025
<i>Alireza Javanmardi⁺, Soroush H. Zargarbashi⁺, Aleksandar Bojchevski, Willem Waegeman, Eyke Hüllermeier</i>	
A Multi-Modal Feature Embedding Approach to Diagnose Alzheimer Disease from Spoken Language [Link to paper] 	2019
<i>Soroush H. Zargarbashi, Bagher BabaAli</i>	

Awards

- **Best Poster Award**, ICLR 2026 Workshop: “Catch, Adapt, and Operate: Monitoring ML Models Under Drift” for **CATS: Conformalized Adaptive Test-time Scaling**

Education

Ph.D. University of Cologne , Artificial Intelligence	2022 – Present
• Research Focus: Trustworthy AI — Uncertainty Quantification, Conformal Prediction, and Robustness • Advisor: Prof. Dr. Aleksandar Bojchevski • Selected Publications: ICML 2024, ICLR 2025, NeurIPS 2025 (under review)	
M.Sc. University of Tehran , Artificial Intelligence	2017 – 2020
• Thesis: Speech Enhancement under Adverse Acoustic Conditions • Supervisors: Dr. Bagher BabaAli and Prof. Dr. Yu Tsao (Academia Sinica) • Relevant Graduate Courses: Approximation Algorithm (19.5/20), Data Science (19.25), Advanced Algorithms(17.25/20), Advanced Artificial Intelligence (18.48/20), Machine Learning (17/20), Graph Theory (16.78/20)	
B.Sc. Isfahan University of Technology , Computer Engineering	2012 – 2017
• Thesis: Normalized Dot Product Representation of Graphs	

Academic Services

Academic Reviewing and Committee Work

2022 - Present

- Reviewer: *NeurIPS* (2024, 2025), *ICLR* (2025), *ICML* (2025)
- Program Committee member for *Epistemic Intelligence in Machine Learning Workshop at EurIPS 2025*.

Invited Talks & Selected Presentations

- **Log Meetup - Aachen:** Conformal Prediction for Graphs
- **Algorithmic Earth System Science Workshop:** Conformal Prediction in Noisy Conditions
- **SPP2298 Annual meeting:** Conformal Prediction Graph-aware and Robust
- **MLBoost Reading Group:** Robust Yet Efficient Conformal Prediction Sets
- **Reading Group MLSec:** On Robust Conformal Prediction: Single Certificate and Single Sample

Attended Academic Events

2022 – 2024

- **CISPA Trustworthy AI Summer School (2022):** Robustness, privacy, and fairness of ML models.
- **LoG Learning on Graphs Meetups (2023–2024):** Presented poster (*EvA: Evolutionary Attacks on Graphs*).

Teaching Assistant

University of Cologne (UzK),
and Isfahan Univ. of Tech
(IUT)

- University of Cologne: *Advanced Machine Learning, Elements of Machine Learning*
Co-designed course material, tutorials, and exams
- Isfahan Univ of Tech: *Algorithm Design*, and *Data Structures*.
Tutorials

Lab Instructor / Tutor

IUT — Isfahan University of
Technology

- Taught *Advanced Programming* for 2 years, for undergraduate students through hands-on lab sessions.

Skills

Machine Learning & AI: Graph Neural Networks (GNNs), Large Language Models (LLMs), Multimodal Learning (Audio–Text), Uncertainty Quantification, Conformal Prediction, Adversarial Machine Learning, Epistemic vs Aleatoric Uncertainty, Bayesian Methods, Ensemble Models

Machine Learning & AI: Uncertainty Quantification, Conformal Prediction, Robustness under Distribution Shift, Adversarial Machine Learning, Graph Neural Networks (GNNs), Large Language Models (LLMs), Multimodal Learning (Audio–Text), Bayesian Inference, Ensemble Methods, Test-Time Adaptation

Deep Learning Frameworks: PyTorch (Lightning, DistributedDataParallel), Hugging Face Transformers, PyTorch Geometric, TensorFlow, scikit-learn

Programming & Systems: Python, C++, Bash, Docker, REST APIs, Git

References

Prof. Dr. Aleksandar Bojchevski — Professor, University of Cologne (Trustworthy AI)
Email: a.bojchevski[at]uni-koeln[dot]de | Web: abojchevski.github.io 

Dr. Bagher BabaAli — Associate professor, University of Tehran, School of Math., Stats. & CS
Email: babaali[at]ut[dot]ac[dot]ir | Web: [UT Faculty Page](#) 

Dr. Matthias Sperber — Apple, Machine Translation. Email: m_sperber[at]apple[dot]com